MRA data process – PO Governance

SAP Merchandise Requests: morning file, live changes from SAP (pending until the day's file), and how open MRAs take capacity · v1.1.7 rules, schema v7 · 6 Nov 2026

A · Morning MRA file (Planner, Data Setup)

Planner uploads the MRA workbook (.xlsx / .xlsb) in the Data Setup sidebar

Item-lines sheet and header sheet found by their headers?

- No → Refused – nothing changed (message names the missing columns)
- Yes →

Read item lines: DOC.NO, Generic Article (last 10 digits), Color, TOTAL PO QTY added up per MRA, DEL. DATE, LEVEL · header sheet: PO, status, created by

Normalise level / date / qty – rows that cannot be read are skipped and counted in the summary

Keep a decision SAP sent earlier when the file has none for that MRA

ONE transaction (PRO_SAP_MRA UPLOAD):
1. SAP_MRA_FILE emptied and filled from the file
2. Changes already applied today re-tagged to the new version
3. Pending SAP changes moved into SAP_MRA_FEED (arrival order, Data_Version = this Upload ID)
4. SAP_MRA_FEED_PENDING emptied
5. IsDailyDataUpload = 1, Data_Version = Upload ID

Summary under the slot: MRAs per level, counted now, rows skipped, "N SAP changes received before this file were applied on top of it"

at the upload: pending changes are moved onto the new file (step 3)

B · Live changes from SAP (all day)

SAP sends POST /api/sap/v1/mra-events or /po-events (one message or {"messages": [...]}, max 5,000)

Technical user (HTTP Basic), IP allow-list, interface on – else 401 / 503

Read each message: mra_no, level (1–9, X) – optional action_by, remark, po_no, po_status, occurred_at, msg_id, decision fields

Message understood?

- No → Stored as "parked" with the reason
- Yes →

Today's MRA file uploaded? (IsDailyDataUpload)

- No → NO → temporary table SAP_MRA_FEED_PENDING answer: "pending" (applied when the file is uploaded)
- Yes → YES → SAP_MRA_FEED with the current Data_Version – applied at once

Replay file + messages of the current version: applied · no change · duplicate (same msg_id) · older (occurred_at before the last change) · parked · stored (PO message) · new MRA needs article, colour, qty, delivery date

Answer to SAP: one result per message, same order (+ counts: applied, pending, parked ...)

C · How the tool uses the MRAs

Every New PO request, SAP /decisions, single concept Recommendation, MRA Queues: current MRA position

Morning file + messages of the current data version replayed → level, qty, PO, history of every MRA

In Article Master and single concept?

- No → Set aside – not counted
- Yes →

Delivery date today or later?

- No → Expired – not counted
- Yes →

Level 1, 2, 4 – or level 5 reached by a live change today with the PO not unreleased?

- No → Not counted (3, 6, 7, 8, 9, X, final earlier, PO unreleased)
- Yes → Counted: qty added to the released POs of its colour and delivery month (a live PO change governs it, switch 2)

Max Permissible = what is still free after the plan, the stock, released POs (F2), open MRAs and today's tool approvals – New PO request card shows "Open MRAs from SAP counted in this colour"

Switch 1 "Tool requests use capacity" ON?

- Yes → ON: request saved, approval counts for the day
- No → OFF: check only – raise the request in SAP; tool approvals not counted

file + feed of the current data version → replay

Where the data is kept (SQL Server, schema v7, procedure PRO_SAP_MRA)

SAP_MRA_FILE	the morning file: one row per MRA (article, colour, qty, delivery date, level, POs)
SAP_MRA_FILE_UPLOADS	one row per upload: file, by, counts, how many pending changes were moved
SAP_MRA_FEED	every live change applied, in order, with its Data_Version (also bad ones, "parked")
SAP_MRA_FEED_PENDING	temporary table: changes received before today's file – emptied by the upload
SAP_MRA_CONTROL	one row: Data_Day, IsDailyDataUpload, Data_Version (= Upload ID of the file in use)

A day with IsDailyDataUpload

00:00	New day: IsDailyDataUpload = No (until today's file)
Morning	SAP changes → pending table (answer "pending")
Upload	File replaces the list; pending moved on top; flag = Yes
Day	SAP changes applied at once (same data version)
Re-upload	Same-day changes kept on the new version